Chapter 4: Monte Carlo Methods

Part 1

Question 1. What are Monte Carlo methods in reinforcement learning? Explain their main purpose and how they differ from Dynamic Programming (DP).

Question 2. Explain the difference between first-visit Monte Carlo and every-visit Monte Carlo methods. How does each method estimate the value of a state?

Question 3. What is the difference between state-value functions and action-value functions? Why are action-value functions particularly useful when the environment model is unknown?

Question 4. Explain Monte Carlo control and the role of Generalized Policy Iteration (GPI) in finding an optimal policy.

Question 5. What is the exploring starts assumption? Why is it important for Monte Carlo control, and what are its limitations?

Question 6. Explain the difference between on-policy and off-policy Monte Carlo methods. Provide an example of each.

Question 7. What is importance sampling? Explain the difference between ordinary importance sampling and weighted importance sampling.

Question 8. What is an ε-greedy policy? Explain how it balances exploration and exploitation in reinforcement learning.

Part 2

1. Write the mathematical definition of the return G_t in an episodic task.
2. Derive the first-visit Monte Carlo estimator for $v_\pi(s)$.
3. Write the sample-average update rule for Monte Carlo prediction.
4. Derive the greedy policy improvement equation:

$$\pi(s) = \arg\max_a q(s, a)$$

5. Explain the policy improvement theorem as applied to Monte Carlo control.
6. Derive the importance-sampling ratio for off-policy prediction:

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{\mu(A_k \mid S_k)}$$

7. Write the ordinary importance-sampling estimator for state-value prediction.
8. Derive the weighted importance-sampling estimator for $V(s)$.
9. Derive the incremental update rule for weighted importance sampling.
10. Explain why ordinary importance sampling may have infinite variance.
11. Derive the decomposition of the discounted return into flat partial returns.
12. Explain how importance sampling on truncated returns reduces unnecessary variance.

Part 3

1. In blackjack, explain how the state is represented and how the reward is defined.
2. Why is it difficult to apply dynamic programming directly to blackjack?
3. How can Monte Carlo methods estimate the value of a blackjack state?
4. Suppose a state is visited five times and the returns are $2, 4, 6, 8, 10$. Calculate its Monte Carlo value estimate.
5. Suppose the returns observed after first visits to a state are $3, 5, 7, 9$. Calculate the first-visit MC estimate.
6. Explain how exploring starts can be implemented in a simulated blackjack environment.
7. Given a set of action values for a state, determine the greedy action and construct the corresponding policy.
8. Explain how an ε-greedy policy selects actions when two actions have equal estimated values.
9. Given target-policy and behavior-policy probabilities for an episode, calculate the importance-sampling ratio.
10. Given a set of returns and importance weights, calculate the ordinary and weighted importance-sampling estimates.
11. Explain why weighted importance sampling may perform better when importance weights are highly variable.
12. Describe how Monte Carlo control can be applied to the racetrack problem.
13. Explain how the racetrack environment represents states, actions, transitions, and rewards.
14. How does random displacement affect the optimal policy in the racetrack problem?
15. Explain how a Monte Carlo control algorithm could learn an optimal racetrack policy from simulated episodes.